ASSIGNMENT REPORT- 6

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)



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INTRODUCTION:

This assignment observes the generation of 2D data using a Gaussian Mixture Model (GMM) and then applying K-means clustering and classification based on distortion. Each of the two classes is modeled using a mixture of three Gaussian components. The data are generated from three different covariance settings: spherical, diagonal and full covariance. The generated points are visualized, divided in training and testing sets in 80:20 ratio and used to study the clustering boundaries and classification performance.

METHODOLOGY:

1. Gaussian Mixture Model

It represents a probability density as a weighted sum of Gaussian component densities. For a 2D observation x, the model is:

$$p(x) = \Sigma_{i=1}^{K} \pi_i \, N(x; \mu_i, C_i)$$

where,

K - number of components

π_i - mixture weight of component i

μ_i - mean vector

C_i - covariance matrix

2. Pseudo-Random Number Generation using LCG

Linear Congruential Generator is used to produce uniform pseudo-random numbers. It uses a = 1,664,525, c = 1,013,904,223, m = 2^{32} and initial seed $X_0 = 42$. A uniform value in [0, 1) is obtained by dividing the updated state by m.

$$X_{n+1} = (aX_n + c) \bmod m$$

3. Gaussian Point Generation

A standard normal pair is transformed using the mean and covariance of the selected component. For a symmetric positive-definite covariance matrix C = [[a, b], [b, c]], the point is generated as:

$$x_1 = \mu_1 + \sqrt{a}\, z_1$$

$$x_2 = \mu_2 + (b/\sqrt{a})z_1 + \sqrt{(c - b^2/a)}z_2$$

This transformation handles the spherical, diagonal, and full covariance matrices

4. Training and Testing Split

The generated data of each class is shuffled and 80% is used for training data and 20% is used for test data. We generate 5000 samples per class. After an 80:20 split each class has 4000 training points and 1000 testing points.

5. K-Means Clustering

Three initial centroids are selected using an initial farthest point initialization procedure. Each training point is assigned to the nearest centroid in terms of squared Euclidean distance. Then each centroid is moved to the mean of the points assigned to it. The steps are repeated until the centroids move by less than a small tolerance, or the maximum number of iterations is reached. K-means models are fitted separately for the two classes.

$$d^2(x, \mu) = (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2$$

6. Classification Using Distortion

For a test point x, distortion is defined as the minimum squared distance from x to any centroid of a class. The point is assigned to the class with the minimum distortion:

$$D_k(x) = \min_j ||x - \mu_{kj}||^2$$

If $D_1(x) < D_2(x)$, the point is classified as Class 1; or else, it is classified as Class 2. The held-out 20% test data is used to calculate accuracy and a confusion matrix.

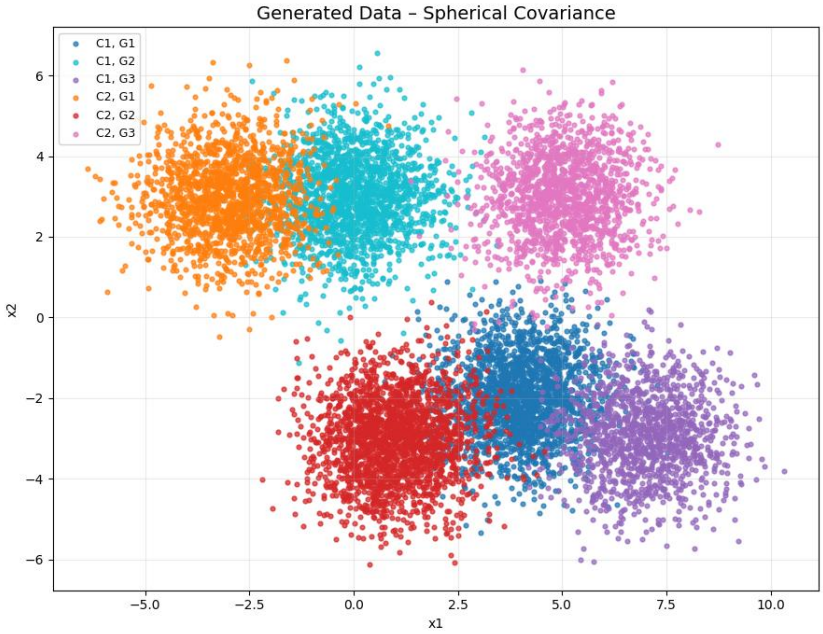
RESULTS AND OUTPUT:

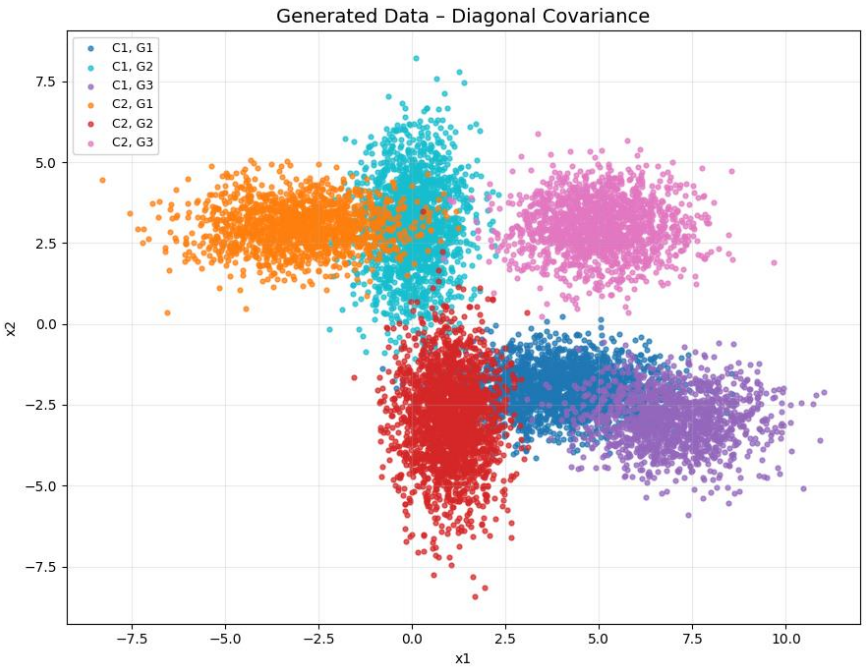
Covariance Type	Test Points (C1 + C2)	Correct Predictions	Accuracy
Spherical	2000	1900	95.00 %
Diagonal	2000	1880	94.00 %
Full	2000	1805	90.25 %

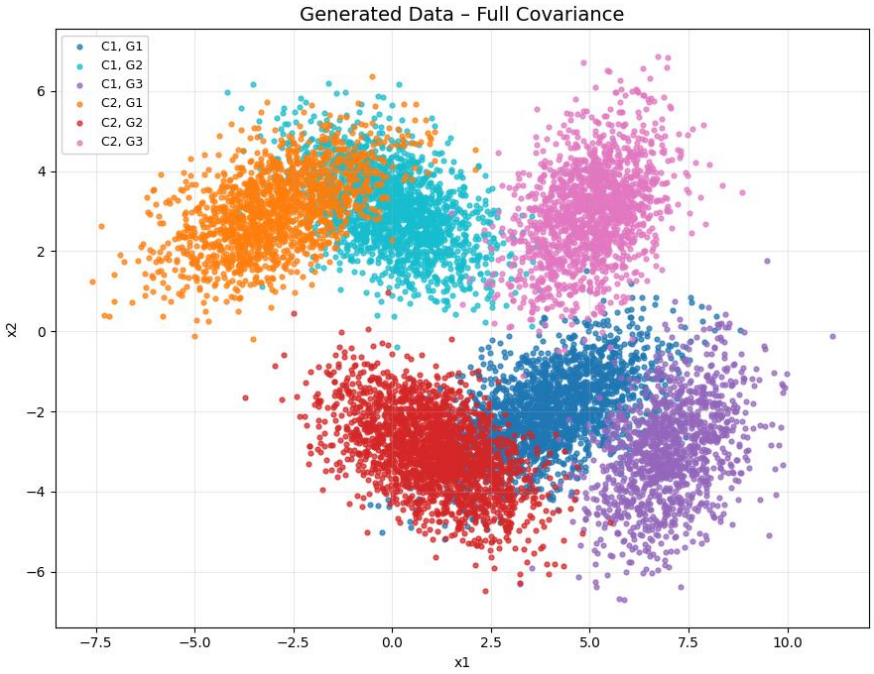
Spherical
Class 1 centroids:
[[3.932, -1.976], [0.035, 3.003], [7.0, -2.947]]
Class 2 centroids:
[[-2.998, 2.963], [5.029, 2.974], [0.999, -3.041]]

Diagonal
Class 1 centroids:
[[6.963, -2.821], [0.008, 3.085], [3.627, -1.993]]
Class 2 centroids:
[[4.991, 3.004], [1.006, -3.013], [-3.035, 3.005]]

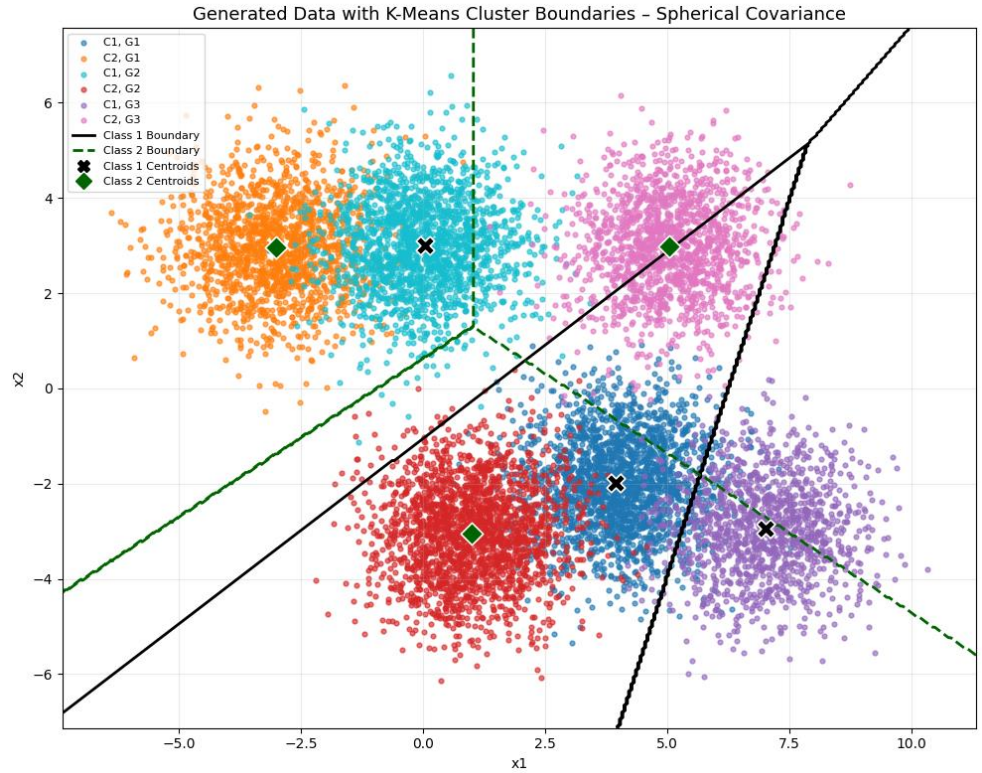
Full
Class 1 centroids:
[[3.491, -2.178], [-0.047, 3.065], [6.811, -2.592]]
Class 2 centroids:
[[5.013, 2.967], [-3.031, 2.982], [1.029, -3.005]]


Generated Data – Spherical Covariance
C1, G1
C1, G2
C1, G3
C2, G1
C2, G2
C2, G3
x1
x2

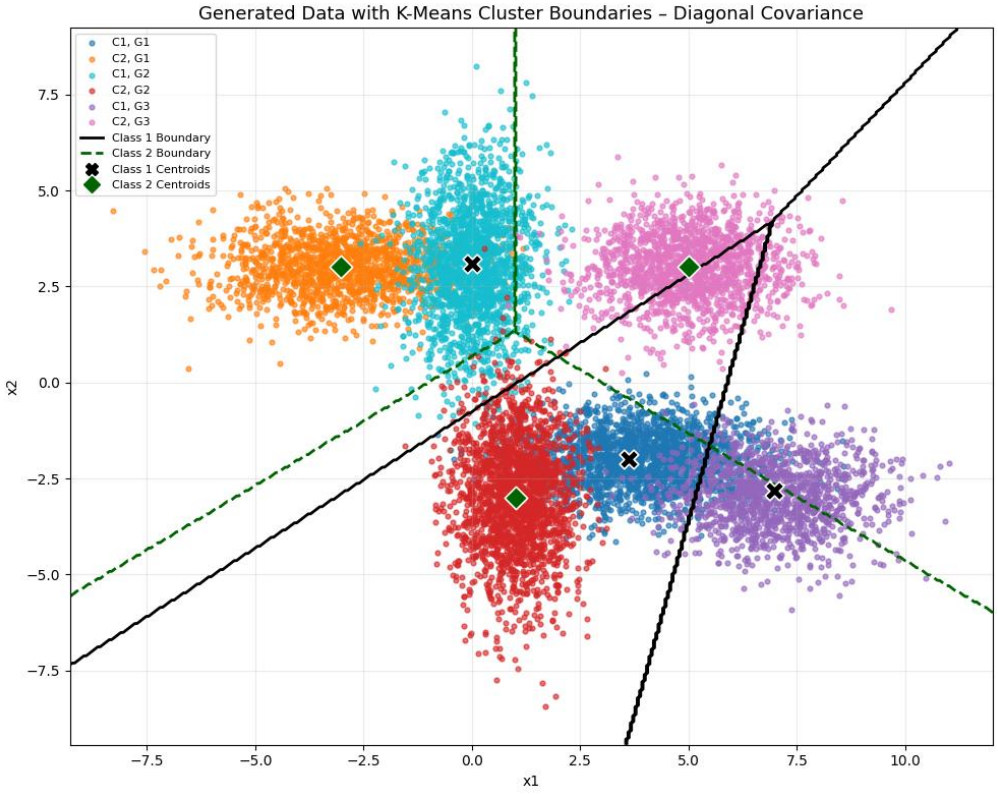

Generated Data – Diagonal Covariance
C1, G1
C1, G2
C1, G3
C2, G1
C2, G2
C2, G3
x1
x2



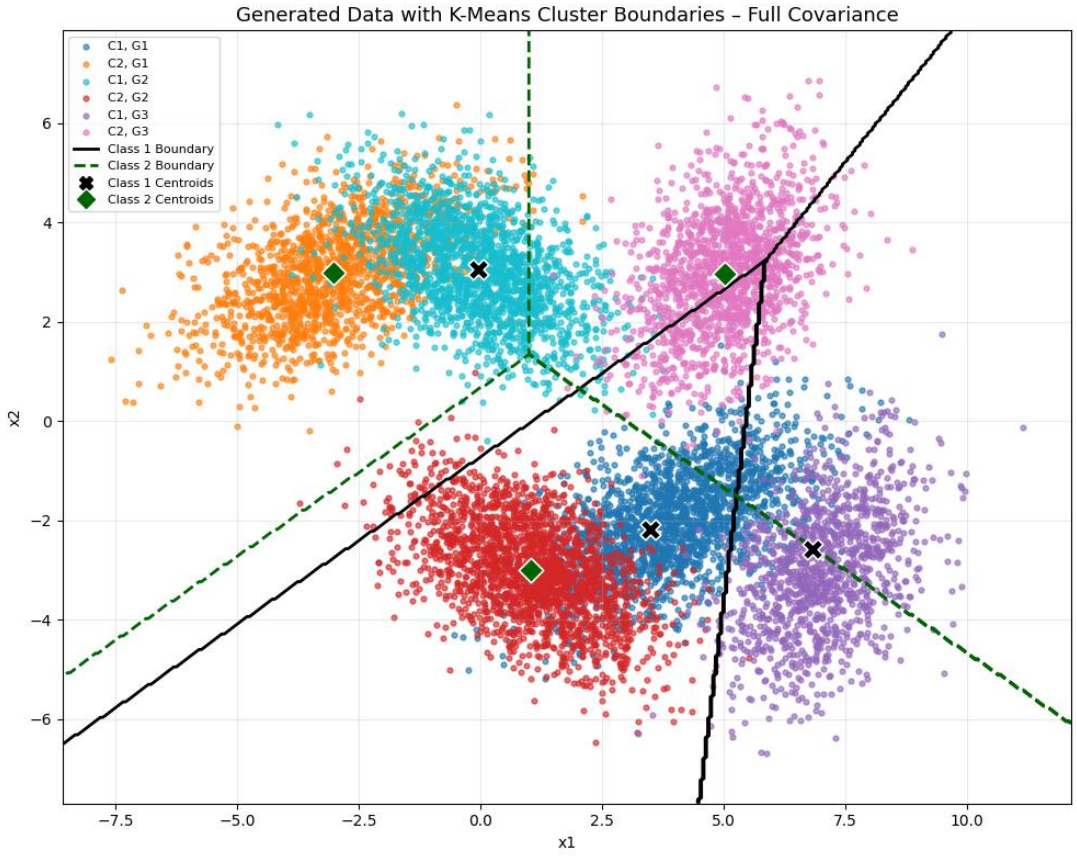
Covariance Type: Spherical
Correct predictions: 1900
Classification accuracy: 95.0 %
Confusion matrix:
[947, 53]
[47, 953]



Covariance Type: Diagonal
Correct predictions: 1880
Classification accuracy: 94.0 %
Confusion matrix:
[951, 49]
[71, 929]



Covariance Type: Full
Correct predictions: 1805
Classification accuracy: 90.25 %
Confusion matrix:
[895, 105]
[90, 910]



Covariance Type: Spherical
Class 1 mean training distortion: 1.9168
Class 2 mean training distortion: 2.0274
Covariance Type: Diagonal
Class 1 mean training distortion: 2.1079
Class 2 mean training distortion: 2.4005
Covariance Type: Full
Class 1 mean training distortion: 2.5087
Class 2 mean training distortion: 2.6327

Conclusion:

This assignment shows the implementation of generating 2D Gaussian mixture data and applying manual K-means clustering for spherical, diagonal, and full covariance. Standard normal values are generated with the Marsaglia polar method and component selection is performed using mixture weights. The training data is used to learn class-specific centroids, while the test data is classified according to the minimum nearest-centroid distortion.

- Classification accuracy is highest for spherical covariance (95.00 %) and decreases for

diagonal (94.00 %) and full (90.25 %) covariance.

- This decline occurs because elongated and rotated clusters produce greater overlap in the Euclidean nearest-centroid decision regions.

- Mean training distortion also increases in the same order (Spherical→ Diagonal-→ Full), which is consistent with the greater geometric complexity of the clusters.

- The visualizations clearly show how the shape of the covariance matrix affects both the data distribution and the Voronoi boundaries produced by K-means.